



Communication and Computer Network CNS3200

Learning outcome

- Identify components in data communication (C4)
- Topics
 - Introduction
 - Data communication
 - Components
 - Example of Protocol
 - Data Representation
 - Transmission Mode
 - Network

Introduction

Why Data Communications?

- accessibility – to get accurate information.
- data/information sharing?
- Eg: From one pc – data copied onto a floppy disk and physically reloaded to another pc/remote computer – time consuming, inconveniences.
- Revolution is occurred in telecommunications networks
- Technological advances drive communication links to carry more and faster signals.

Data Communication

- **Definition:**

1. **Telecommunication**= means communication at a distance (tele is Greek for far). E.g: telephony, telegraphy, and television.

2. **Data** =

- Facts, concepts, and instructions presented in whatever form is agreed upon by the parties creating and using data.
- In the context of computer IS, data are represented by binary information units (or bits) produced and consumed in the form of 0s and 1s.

- **Data communication**

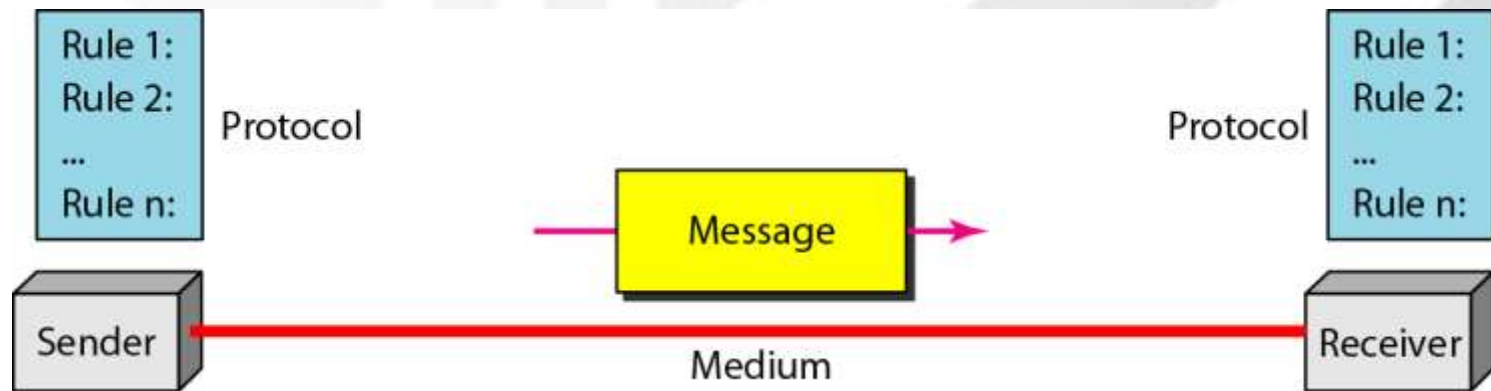
The exchange of data between two devices via some form of transmission medium (such as wire cable)

Data Communication

The effectiveness of a data communication system depends on 4 fundamental characteristics :

- **Delivery**
The system must deliver data to the correct destination.
- **Accuracy**
The system must deliver data accurately.
- **Timeliness**
The system must deliver data in timely manner.
- **Jitter**
Variation in arrival time.

Data Communication



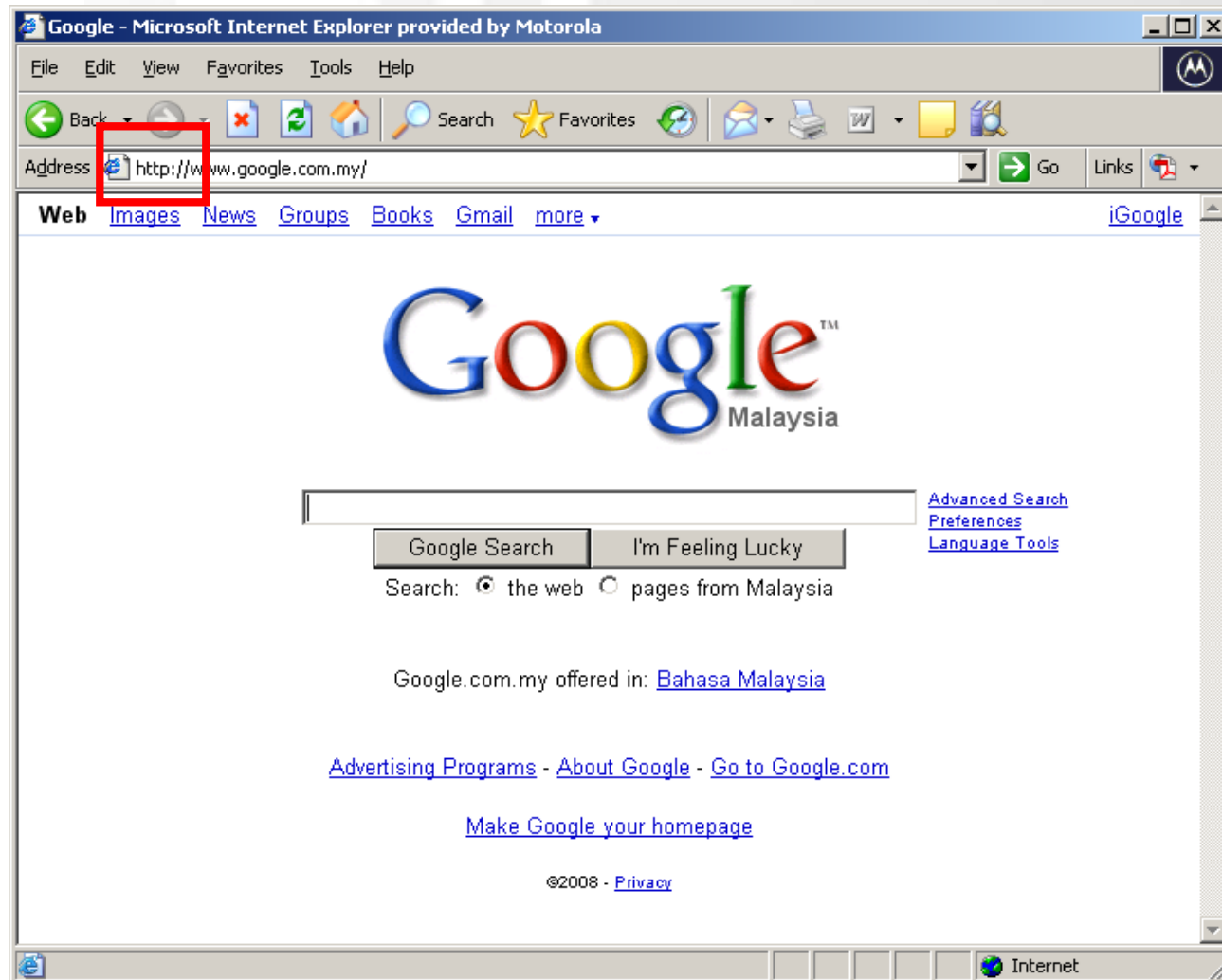
Data communication components

Data Communication Components

A data communication system consists of 5 components :

1. **Message**– The message is the information (data) to be communicated. Can consists of text, numbers, picture, audio, video.
2. **Sender** – the device that sends the data message. Can be a computer, workstation, mobile phone, video cam etc
3. **Receiver** – the device that receives the message. Can be a computer, workstation, mobile phone, tv etc.
4. **Medium** – the physical path by which a message travels from sender to receiver. UTP cable, coaxial, fiber optic, radio wave.
5. **Protocol** – A set of rules that govern data communication. Represent an agreement between the communicating devices.

Example of Protocol



Data representation

Types of data representation

1. Text

In the bit format (“0”, “1”) – bit pattern

Known as code – a set of bit sequence

le: - Unicode

2. Numbers

Also represented by bit patterns

3. Images

Also represented by bit patterns associated with each pixels.

le: black pixel – all ‘1’s

4. Audio

1. Video

Transmission mode

- **Term transmission mode** – the direction of signal flow between two linked devices.
- **There are three types:**
 - Simplex
 - Half-duplex
 - Full-duplex

Transmission mode

- **Simplex**

- The communication is **unidirectional**. Only one of two stations on a link can transmit; the other can only receive
- E.g: keyboards and traditional monitor

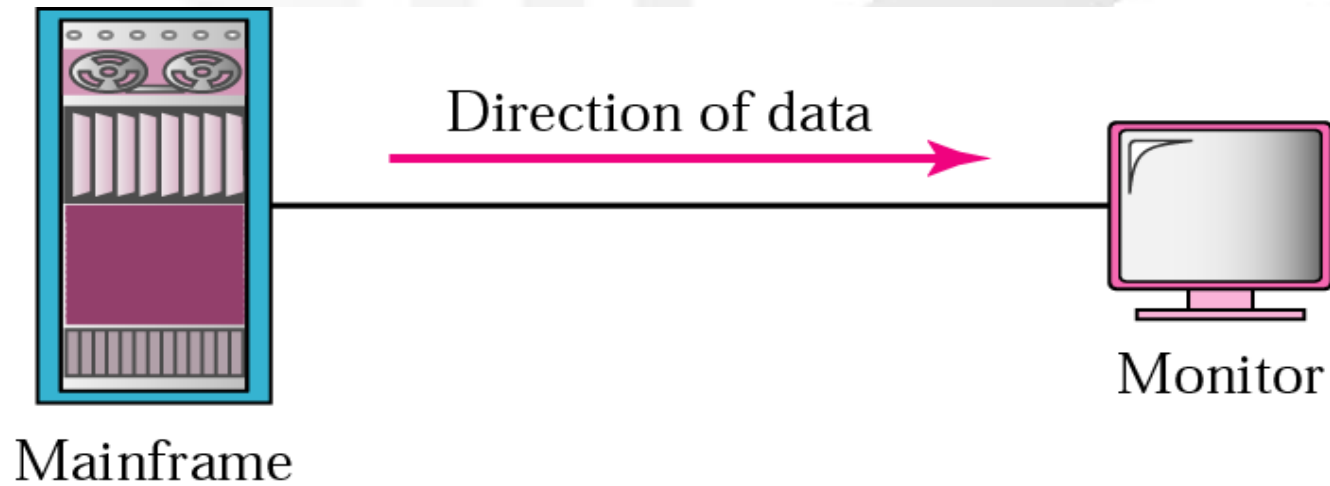
- **Half-Duplex**

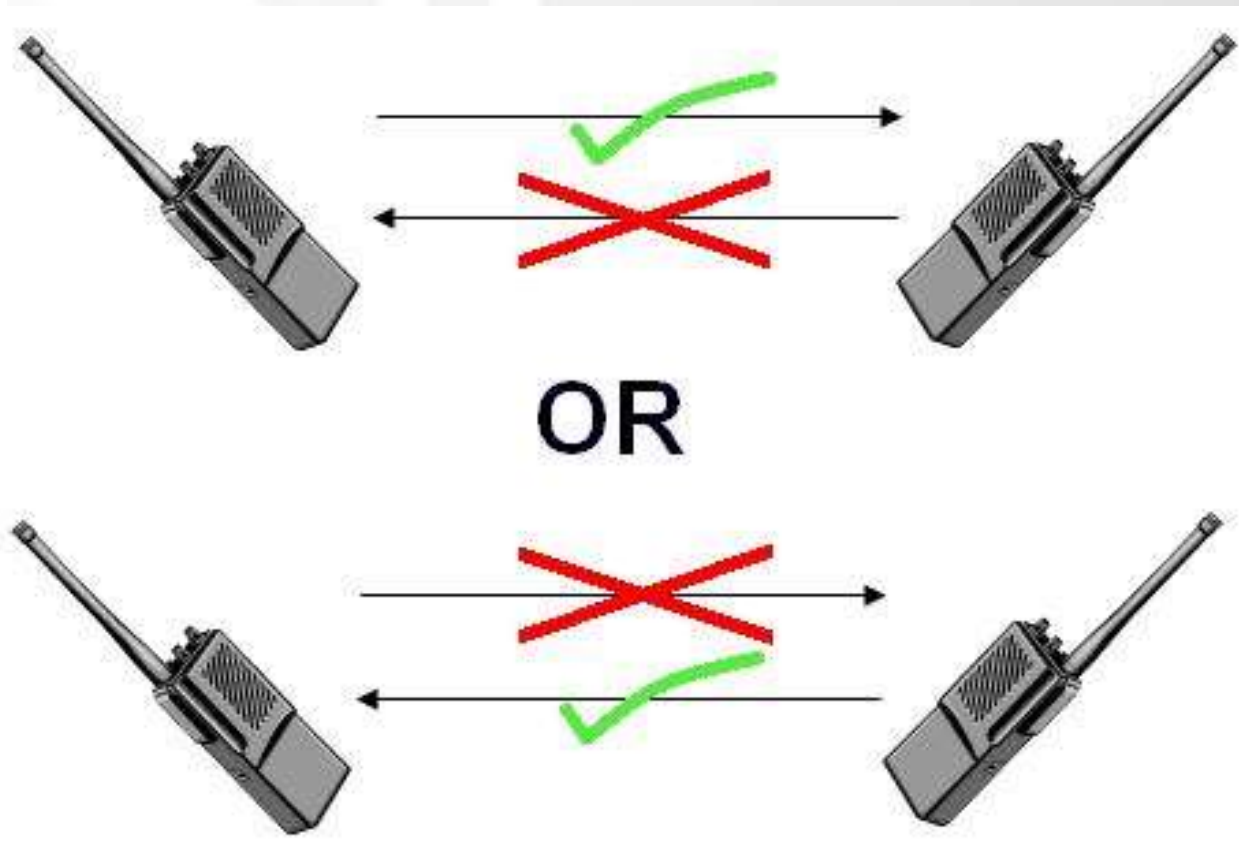
- Each station can **both transmit and receive**, but **not at the same time**.
- When one device is sending, the other can only receive, and vice versa
- The entire capacity of a channel is taken over at the time
- E.g: walkie-talkie

- **Full-Duplex**

- Both stations can transmit and receive simultaneously.
- Like two way street, traffic flowing in both directions at the same time.
- Sharing the capacity of the link.
- E.g: telephone line, videoconferencing

Simplex

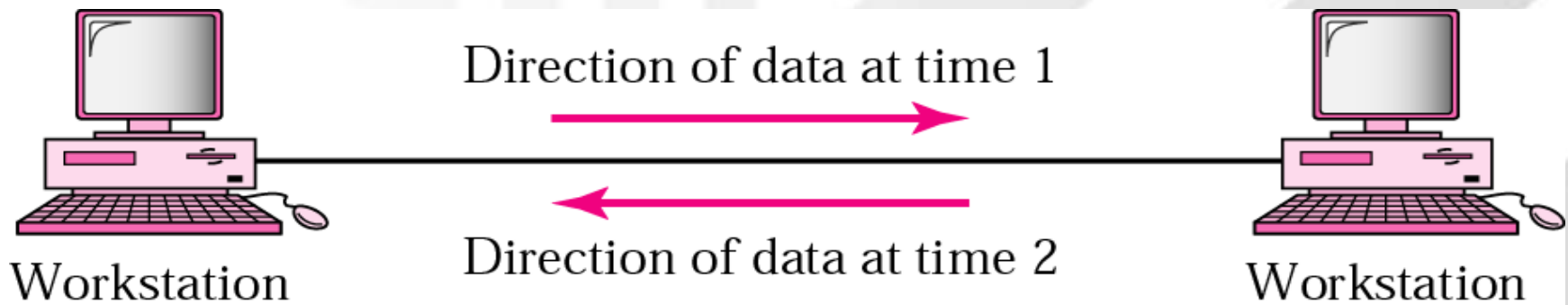




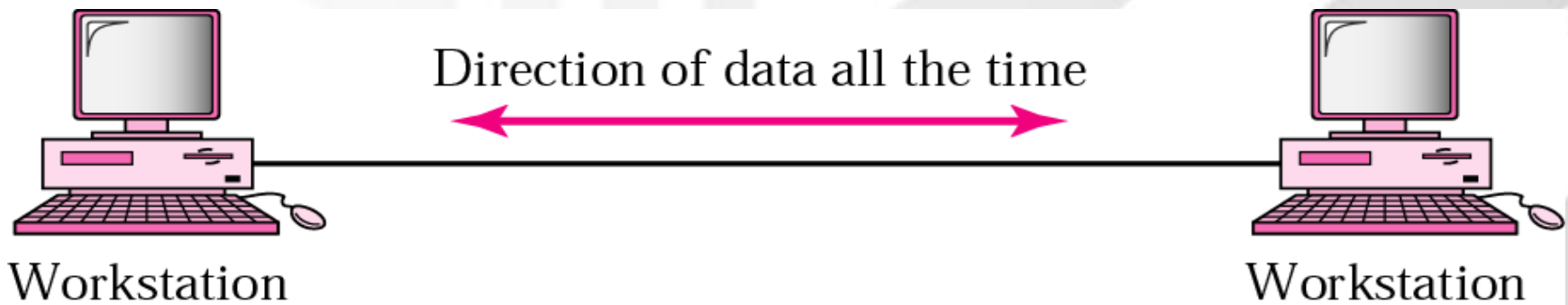
Half-Duplex

(Only 1 communication at a single time can be transmitted)

Half-Duplex



Full-Duplex



Networks

- A **network** is a set of devices (often referred to as **nodes**) connected by communication **links** (channel).
 - A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.
- Networks use **distributed processing**, in which task is divided among multiple computers
- Network criteria:
 - Performance – ie: transmit time and response time
 - Reliability – frequency of failure
 - Security – protecting data from unauthorized access and damage.



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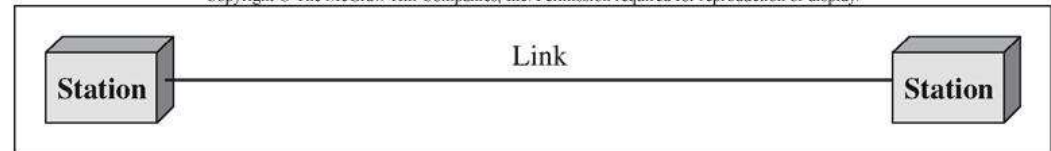
Learning outcome

- Illustrate/draw network topology and architecture (C4)
- linking the importance of protocols and standards (C3)
- Topics
 - Topology
 - Network models
 - Protocol and standard

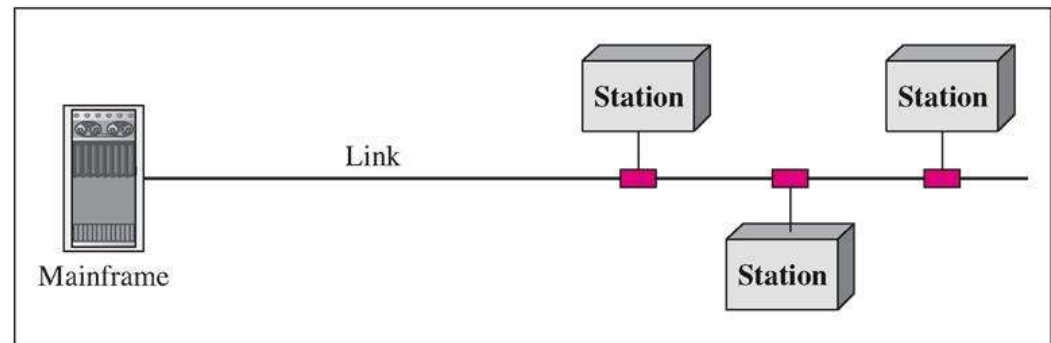
Physical Structure

- Type of connection
 - Point-to-point
 - Multipoint
- Physical Topology

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a. Point-to-point



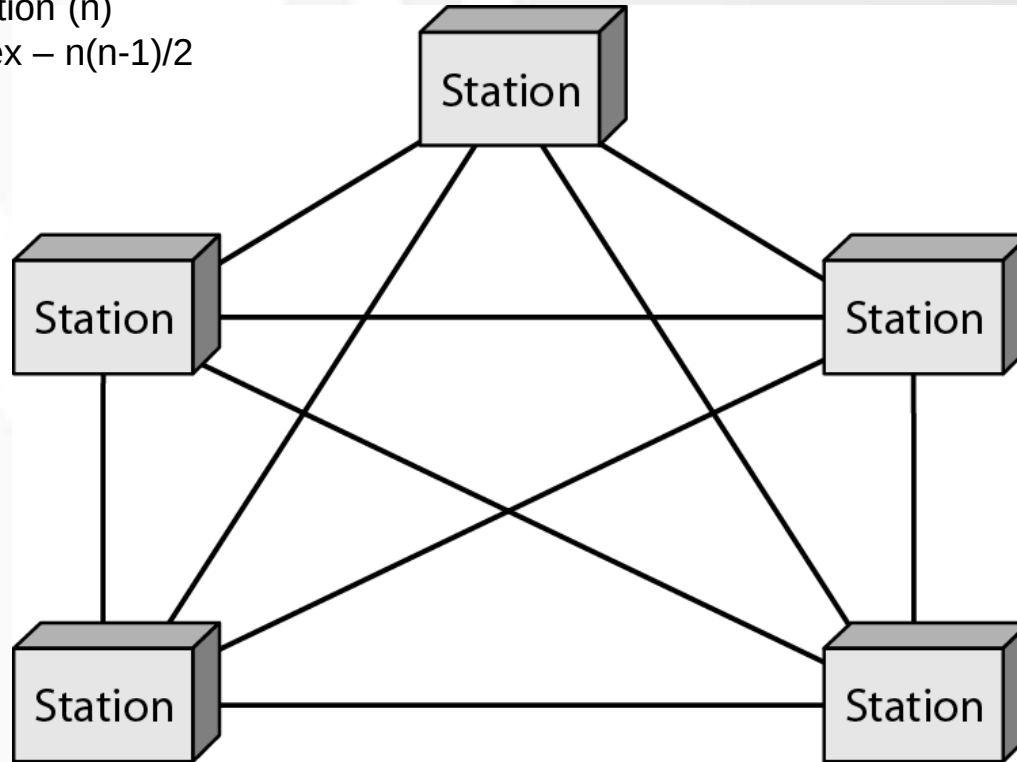
b. Multipoint

Physical Topology

- Physical topology – refers to how the network is laid out physically
- 4 basic topology
 - Mesh - dedicated point-to-point link to every other device (see figure 1.5 in the text book)
 - Star - dedicated point-to-point link **only** to a central controller (see figure 1.6 in the text book)
 - Bus – all devices are linked through a long cable acts as a backbone (see figure 1.7 in the text book)
 - Ring – dedicated point-to-point to its adjacent devices (see figure 1.9 in the text book)
- Hybrid topology
 - A collection of any two or more basic topology
 - Ie: a star backbone + three bus network (fig 1.9)

Nodes/Station (n)
> full duplex – $n(n-1)/2$

$$\frac{5(5-1)}{2} = 10 \text{ links}$$



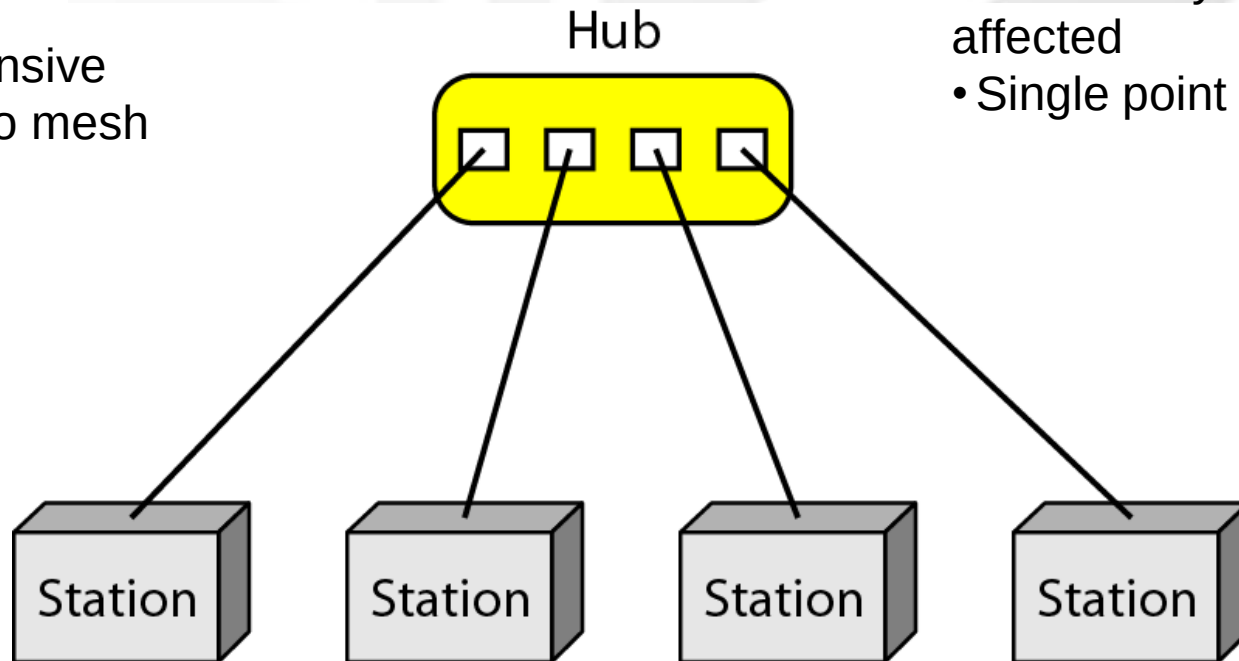
Require a
dedicated link

A fully connected mesh topology (five devices)

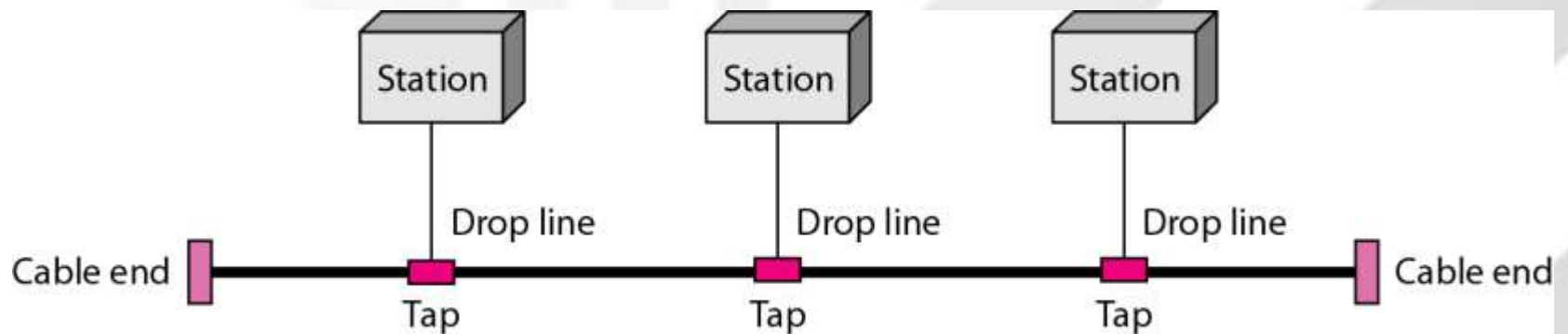
- Central controller
- All communication are through the hub
- Is used in Local-Area Network (LAN)

- Less expensive compared to mesh topology

- Robustness – when a link fail- only the link is affected
- Single point of failure



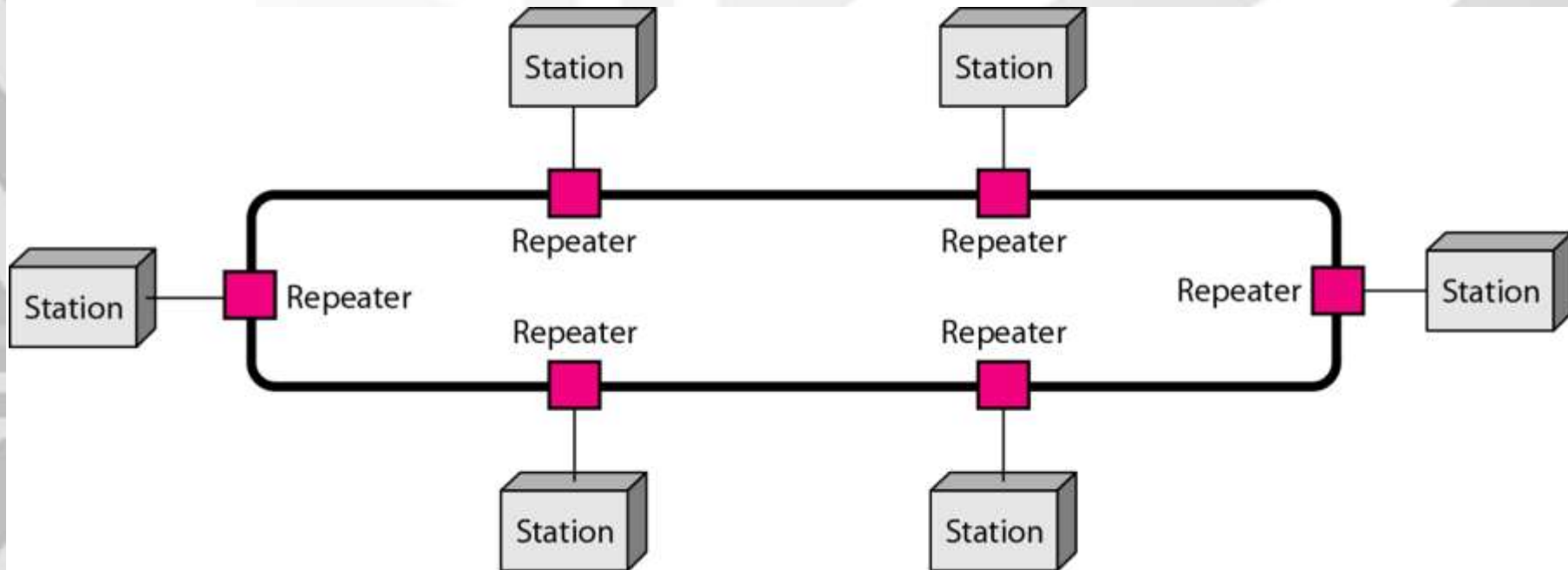
A star topology connecting four stations



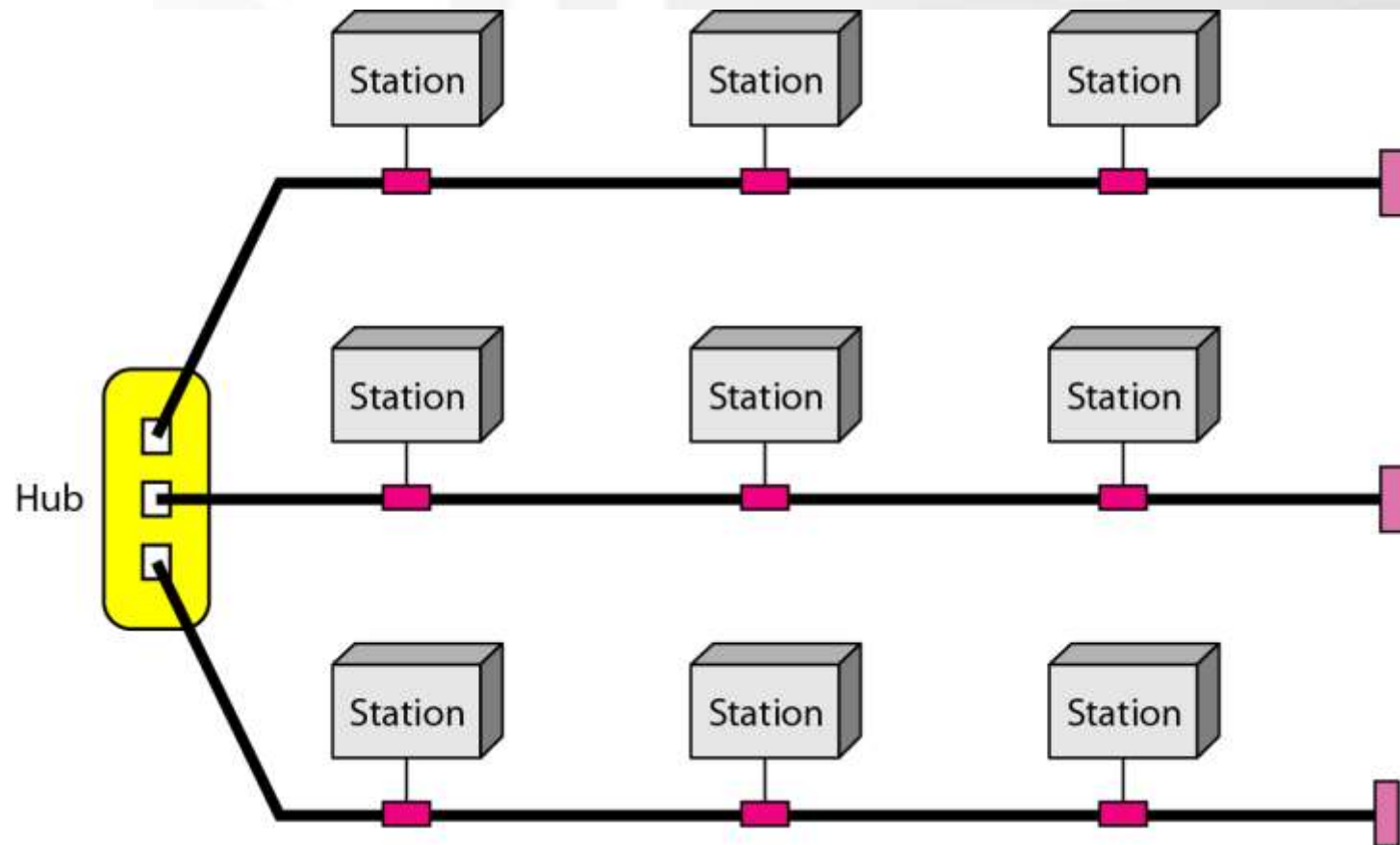
A bus topology connecting three stations

- Connected by drop line and tap
- Signal travel through cable = energy are converted to heat
- Therefore it become weaker with the distance

- Connected to its immediate neighbours
- Add/delete device requires two connections



A ring topology connecting six stations



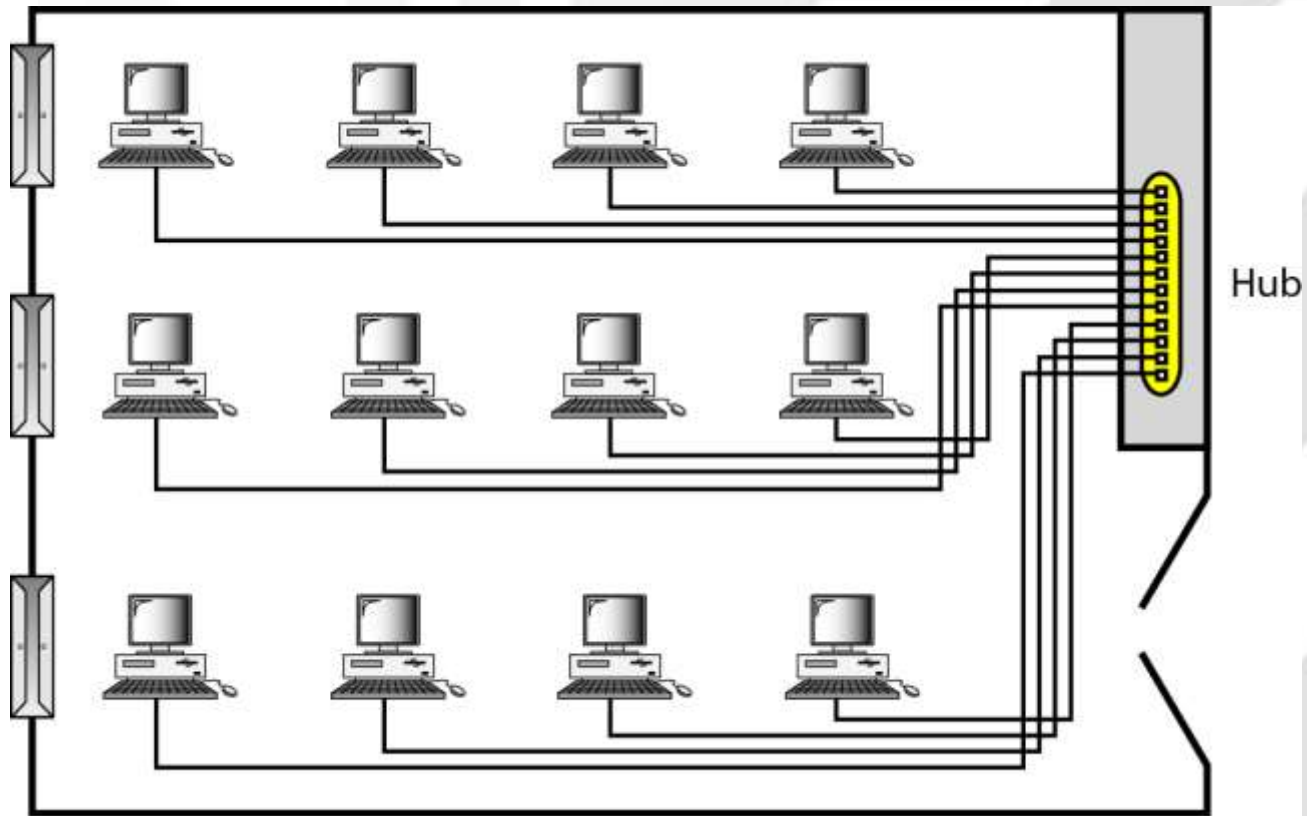
A hybrid topology: a star backbone with three bus networks

Categories of Networks

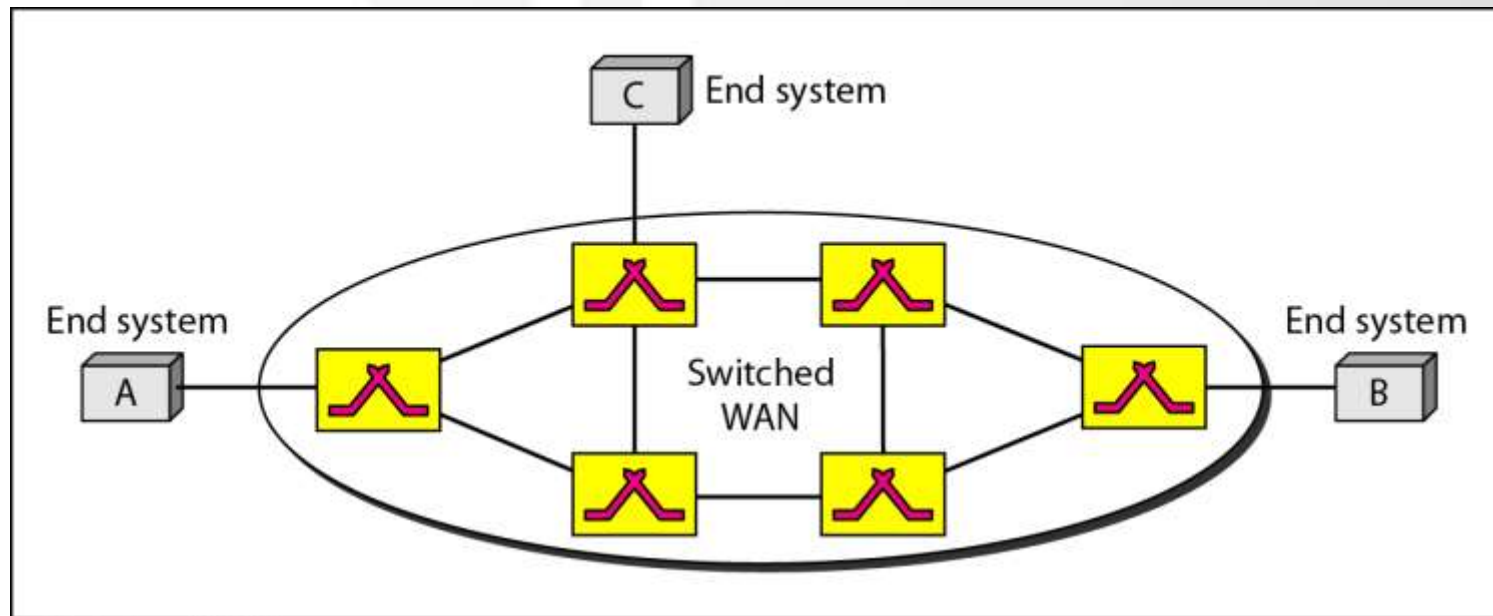
- Is based on size, ownership, the distance it covers and physical architecture.
- LAN
 - Privately owned
 - Link the devices in a single office, building, or campus
 - Limited to a few kilometer
 - **Resource sharing** between PC/workstation
 - Other factor: transmission media and topology
 - Most common topologies: bus, ring, star
 - Data rate/speed up to 100Mbps
- MAN
 - Designed to extend over an entire city
 - Ex: cable TV network, LAN to LAN resource sharing
 - Operates as private company or public company -maxis26

Categories of Networks

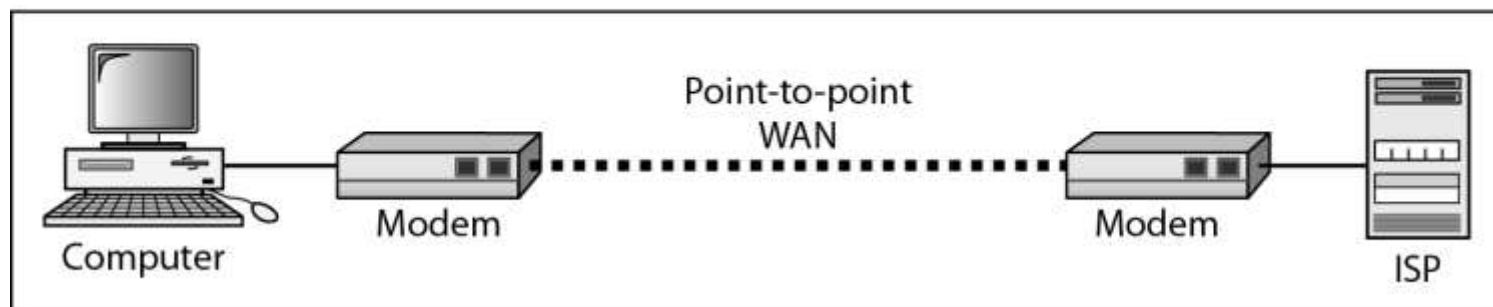
- WAN
 - Provides long distance transmission of data, voice, image, video conference over large geographic areas
 - May utilize as public, leased or private communication equipment.
 - Enterprise network – refer to WAN that wholly owned and used by a single company.



An isolated LAN connecting 12 computers to a hub in a closet

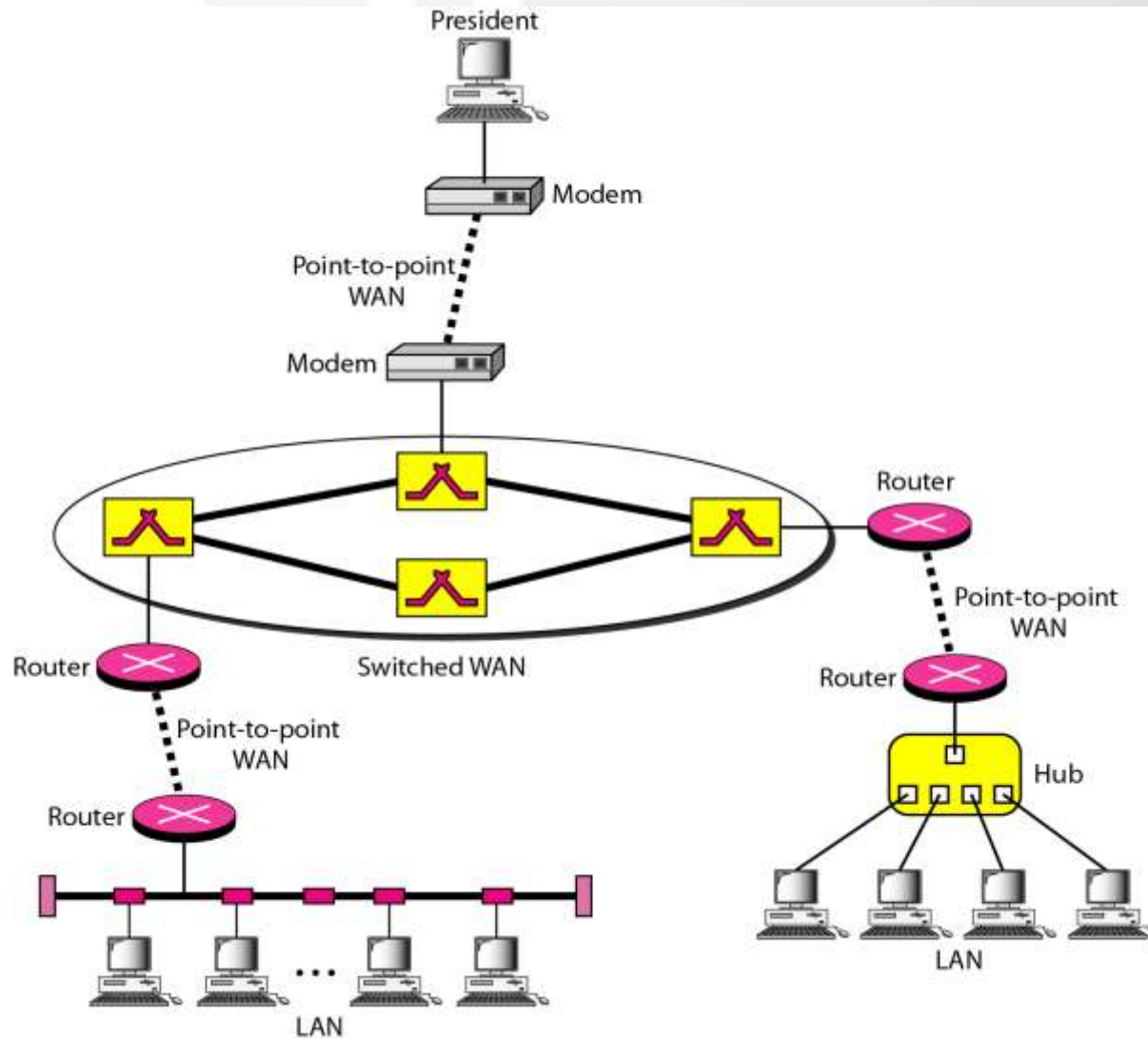


a. Switched WAN



b. Point-to-point WAN

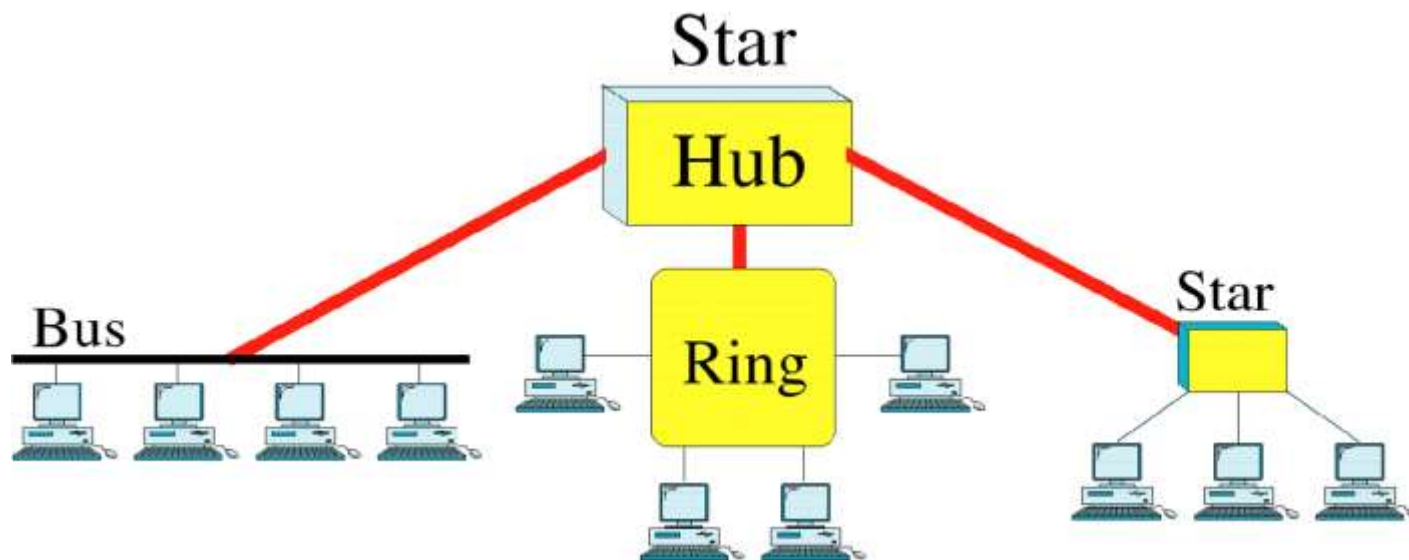
WANs: a switched WAN and a point-to-point WAN



A heterogeneous network made of four WANs and two LANs

Topology

- **Hybrid Topologies**
- Often a network combines several topologies as subnetworks linked together in a larger topology
- Several topology can be connected to each other via a central controller in a star topology





```
TRACERT
Auto
Microsoft(R) Windows 98
(C)Copyright Microsoft Corp 1981-1998.
C:\Windows\Desktop>tracert www.fu.is.saga-u.ac.jp
Tracing route to lark.fu.is.saga-u.ac.jp [133.49.23.131]
over a maximum of 30 hops:
  1      1 ms    <10 ms    <10 ms    202.184.29.252
  2      1 ms    <10 ms    <10 ms    202.184.17.254
  3      1 ms    1 ms      1 ms      202.184.41.2
  4      1 ms    2 ms      2 ms      161.142.158.1
  5      1 ms    1 ms      1 ms      61.6.41.1
  6      1 ms    2 ms      2 ms      161.142.23.169
  7      5 ms    7 ms      6 ms      pos0-0.mea90.jaring.my [161.142.25.86]
  8     191 ms   191 ms    192 ms    pos1-0.tlg90.jaring.my [61.6.17.34]
  9     191 ms   192 ms    191 ms    ix-0-1-2-0.har1.LosAngeles.teleglobe.net [66.110
.10.9]
 10     192 ms   191 ms    191 ms    if-2-1.core1.LosAngeles.Teleglobe.net [207.45.22
0.97]
 11     191 ms   191 ms    191 ms    ix-6-0.core1.LosAngeles.Teleglobe.net [207.45.19
3.82]
 12     194 ms   194 ms    193 ms    205.171.4.186
 13     217 ms   218 ms    218 ms    203.181.106.54
 14     216 ms   217 ms    217 ms    kyotoS1-G0-1.sinet.ad.jp [150.99.198.209]
 15     220 ms   218 ms    218 ms    kyoto-11-P5-0.sinet.ad.jp [150.99.197.242]
 16     320 ms   251 ms    241 ms    kyoto-10-G2-0.sinet.ad.jp [150.99.199.102]
 17     237 ms   250 ms    237 ms    kyushu-10-A5-0-1.sinet.ad.jp [150.99.161.6]
 18     241 ms   241 ms    *         kyushu-1-A3-0-1.sinet.ad.jp [150.99.198.101]
 19     382 ms   270 ms    261 ms    saga-u.gw.sinet.ad.jp [150.99.208.18]
 20     281 ms   314 ms    268 ms    133.49.250.1
 21     242 ms   241 ms    252 ms    133.49.200.253
 22      *      *         *         Request timed out.
 23      *      *         *         Request timed out.
 24      *      *         *         Request timed out.
 25      *      *         *         Request timed out.
 26      *      *         *         Request timed out.
 27      *      *         *         Request timed out.
```

FSKTM, UPM



IDEC, UPM



JARING MIMOS
(MALAYSIA)



TELEGLOBE
(LOS ANGELES, USA)



KYOTO, JAPAN



KYUSHU, JAPAN

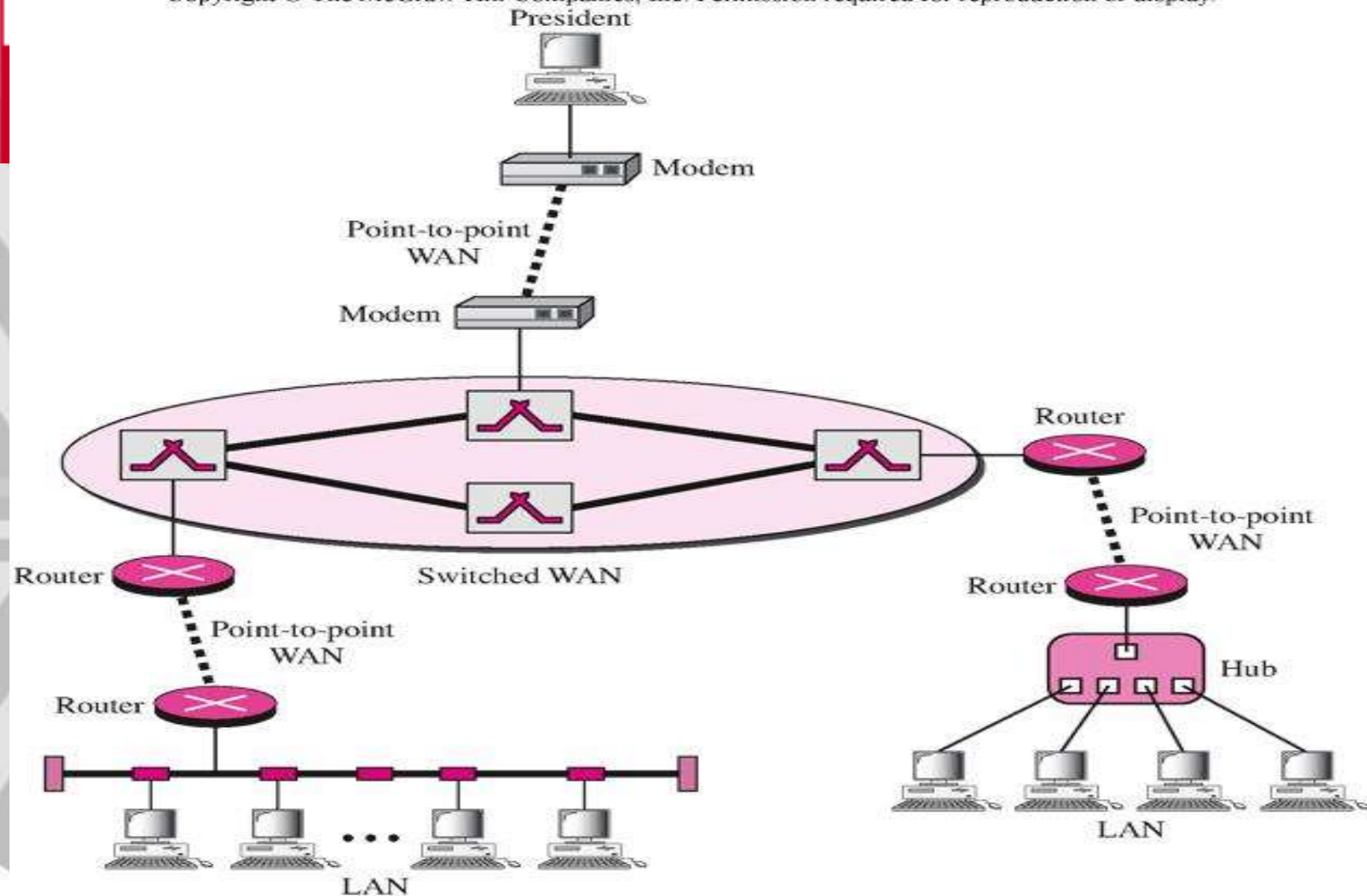


SAGA, JAPAN

Internetworks

- **Internetwork/internet** – two or more networks are connected
- **Internetworking devices** – routers, gateways
- **internet** – a generic term used to mean an interconnection of networks
- **Internet** – a specific worldwide network

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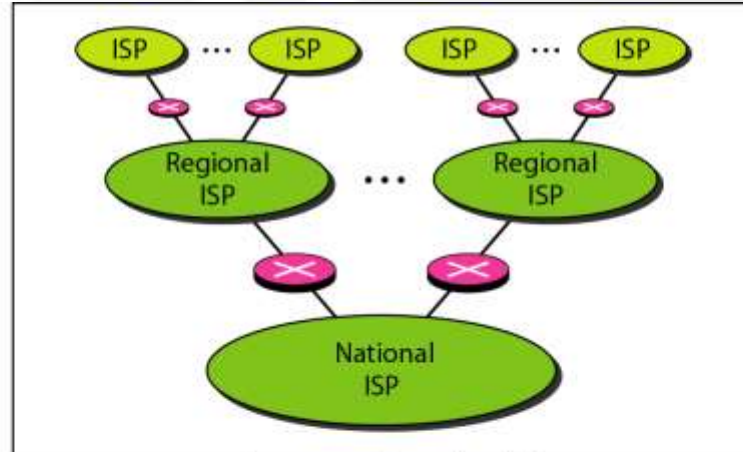
The Internet

The **Internet** has revolutionized many aspects of our daily lives. It has affected the way we do business as well as the way we spend our leisure time. The Internet is a communication system that has brought a wealth of information to our fingertips and organized it for our use.

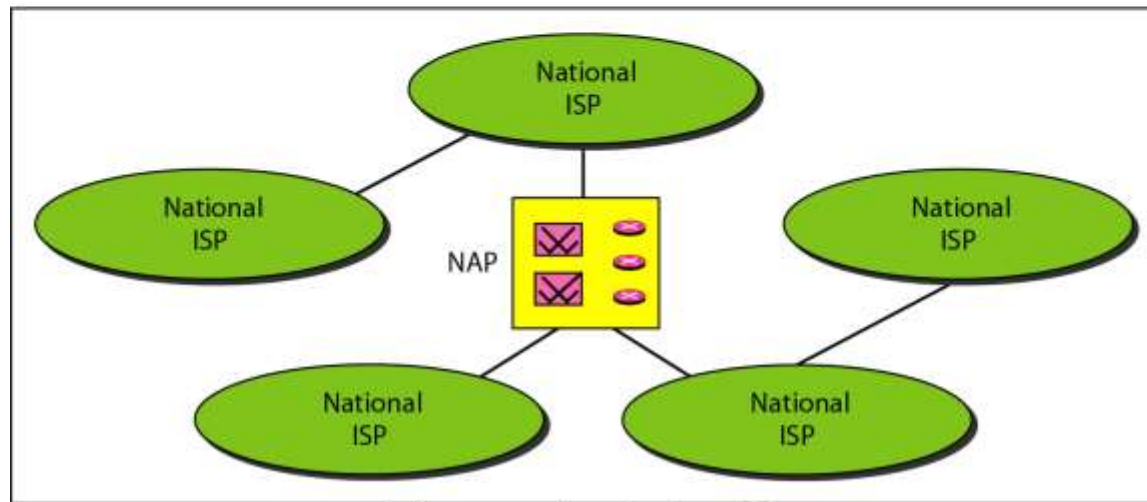
Topics discussed in this section:

A Brief History

The Internet Today (ISPs)



a. Structure of a national ISP



b. Interconnection of national ISPs

Hierarchical organization of the Internet

Protocols and Standards

In this section, we define two widely used terms: **protocols** and **standards**. First, we define protocol, which is synonymous with rule. Then we discuss standards, which are agreed-upon rules.

Topics discussed in this section:

Protocols

Standards

Standards Organizations

Internet Standards

Protocol & Standard

- **Protocol** – set of rules that govern data communication.
 - For the simplicity -the protocol is the collection of rules- definition on what, how and when it is communicated.
- Key element of a protocol are:
 - **Syntax** – refers to the structure or format of the data, the order in which they are presented.
 - **Semantics** – refers to the meaning of each section bits.
 - **Timing** – refers to two characteristic: when data should be sent, and how fast they can be sent
- **Standard** –provides a model for development that makes it possible for a product to work regardless of individual manufacturer
- When no standards, difficulties arise. Example: automobile

Cont..

- Standard – creating and maintaining an open and competitive market for equipment manufacturers, guaranteeing national/international interoperability of data and telecommunications tech process
- Data com standard can be categorized to 2:
 - **De jure (by law/regulation)**
 - Legislated by an officially recognized body
 - **De facto (by fact/convention)**
 - Proprietary, nonproprietary

Standards Organization

- Standard are developed by cooperation among **standards creation committees, forums, and government regulatory agencies.**
- **Standards Creation Committees:**
 - **ISO** – International Standards Organization
 - **ITU-T** – International Telecommunications Union –Telecommunication Standards Sector
 - **ANSI** – American National Standard Institute
 - **IEEE**- Institute of Electrical and Electronic Engineers
 - **EIA** – Electronic Industries Association
 - Telcorda
- ISO – multinational body whose membership is drawn mainly from the standards creation committees of various governments throughout the world.
 - Aim – to facilitate the international exchange of goods and services by providing models for compatibility

Standard Organization

- **ISO** – International Standards Organization
- This organization contributes to the standardization of large networks, such as Internet standard. ISO developed the OSI model. This model describes the principles of networks, constructing a simple and clear architecture for computer network.
-
- **ITU-T** – International Telecommunications Union –Telecommunication
- ITU is committed to standardizing large telecom networks, such as X.25 and frame relay.
-
- **ANSI** – American National Standard Institute
- This organization defined the standards of FDDI (fiber distributed data interface). It is a voluntary organization.
-
- **IEEE**- Institute of Electrical and Electronic Engineers
- This organization provides standard of network hardware. Based on the standards, hardware manufacturers produce compatible product. IEEE formulated 802. project for LAN including ethernet,token ring,WLAN etc
-
- **EIA** – Electronic Industries Association
- This association defines the standard of network cables, such as cAT5, RS232 etc

Standards Organization

- ITU-T
 - formerly known CCITT – Consultative Committee for International Telecommunication Union
 - Related to UN which developed standards for telecommunications
 - Two popular standards – V series and X series
- ANSI
 - US non-profit organization
 - US representative to both the ISO and the ITU-T
- IEEE
 - The largest professional engineering society in the world
 - Develops standards for computing, communication, electrical engineering and electronics
 - Sponsored an important standard for LAN called Project 802

Standards Organization

- EIA
 - Association of electronics manufacturers in US
 - Responsible for developing the EIA-232-D and EIA-530 standards – define serial transmission between two digital devices. (computer to modem)
- Telcordia
 - Provides R&D resources for the advancement of telecommunication technology.
 - Important source of draft standard to ANSI
- **Forums**
 - Telecommunications technology development is moving faster than the ability of standard committee to ratify standards
 - Standards committee are procedural bodies and by nature slow moving

Standards Organization

- The forums work with **universities** and **users** to test, evaluate and standardize new technologies
- At the end, the forums present their conclusion to standards bodies
- Example – Frame Relay Forum, ATM Forum, Internet Society (ISOC), Internet Engineering Task Force(IETF)
- **Regulatory Agencies**
 - All communication technology is subject to regulation by government agencies such as the Federal Communication Commission (FCC) in US.
- **Internet Standard**
 - Tested specification to be used for Internet
 - Being published in Internet draft as Request for Comment(RFC) document

Examples of ISO standard

ISO - International Organization for Standardization - Microsoft Internet Explorer provided by Motorola

File Edit View Favorites Tools Help

Address <http://www.iso.org/iso/home.htm>

ISO International Organization for Standardization
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ISO 9000 - Selection and use

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- Publications and e-products

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- Conformity assessment
- Developing countries
- Social Responsibility
- Standards and regulations
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News and media

- News
- Magazines
- Hot topics
- The ISO Café
- Events
- Media kit

About ISO

- The ISO system
- Members
- Strategy and policies
- ISO/IEC Information Centre
- Jobs at ISO/CS

Management standards

- Understand the basics
- Benefits
- ISO 9000/ISO 14000

Education and training

- Education
- Technical assistance
- Training

IEEE Specification

Project 802 Specs - Microsoft Internet Explorer provided by Motorola

File Edit View Favorites Tools Help

Back Forward Stop Reload Home Search Favorites Refresh Print Mail New Window

Address <http://www.kinnett.net/networkingtips/project802.html> Go Links

Project 802 Specs

Home Up

IEEE Project 802 Specifications

- 802.1 - Internetworking
- 802.2 - Logical Link Control (LLC)
- 802.3 - Carrier-Sense Multiple Access with Collision Detection (CSMA/CD) LAN (Ethernet)
- 802.4 - Token-Bus LAN
- 802.5 - Token-Ring LAN
- 802.6 - Metropolitan Area Network (MAN)
- 802.7 - Broadband Technical Advisory Group
- 802.8 - Fiber-Optic Technical Advisory Group
- 802.9 - Integrated Voice/Data Networks
- 802.10 - Network Security
- 802.11 - Wireless Networks
- 802.12 - Demand Priority Access LAN, 100BaseVG-AnyLAN